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 SURVEY

The Evolution of VANET: A Review of Emerging Trends in Artificial Intelligence and Software-Defined Networks

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ABSTRACT This paper presents an in-depth look at the evolution of Vehicular Ad Hoc Networks (VANET) and their increasing importance in the development of Intelligent Transportation Systems (ITS). Emerging technologies that shape the future of VANET are discussed, focusing on artificial intelligence (AI) and software-defined networking (SDN) as key solutions to the challenges inherent in these dynamic networks. Applications of AI, including machine learning (ML), deep learning, and swarm intelligence, for improving security, traffic efficiency, and information management in VANET were examined. Routing challenges in these environments are discussed and SDN is presented as a viable solution. SDN protocols, controllers, and architectures are described, and their advantages and disadvantages in the context of VANET are analyzed. It provides a comprehensive overview of the current state of VANET, the emerging technologies driving them, and the challenges and opportunities presented by their continued development. The convergence of AI, SDN, and other key technologies is crucial for intelligent transportation, paving the way for safer, more efficient, and more connected driving.

INDEX TERMS Artificial intelligence, software-defined networking, routing, vehicular ad hoc networks, wireless networks.

I. INTRODUCTION

Vehicular Ad Hoc Networks (VANET) are a specialized type of mobile ad hoc network (MANET) that use vehicles as mobile nodes to create a communication network. These vehicles, equipped with on-board units (OBUs) and sensors, can communicate with each other with road infrastructure such as roadside units (RSUs) and can even feature vehicle-to-pedestrian (V2P), vehicle-to-network (V2N), and vehicle-to-everything (V2X) communications. This continuous exchange of information in

real-time allows VANET to offer a variety of services designed to improve road safety, optimize traffic flow, and provide information and entertainment to drivers and passengers [1], [2].

VANET are an integral part of Intelligent Transportation Systems (ITS) and are instrumental in shaping the future of transportation. They enable the development of applications that warn of collisions, provide information on traffic congestion, offer lane-keeping assistance, facilitate access to entertainment services, such as streaming media content, and enable emergency services. [2], [3], [4], [5]. The importance of VANET lies in their ability to create safer, more efficient, and more comfortable driving environments, thereby

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contributing to the development of smart cities with smoother traffic and fewer accidents.

Despite their potential, large-scale deployment of VANET faces challenges owing to the dynamic nature of the network. High node mobility coupled with unstable connectivity resulting from obstacles and limited communication ranges present difficulties in maintaining smooth communication [6], [7], [8]. Furthermore, the need to manage large volumes of data in real time, ensure information security and privacy, and address network security concerns are critical factors that must be addressed for VANET to achieve their full potential [9], [10], [11].

The move towards smarter transportation systems also involves the integration of emerging technologies such as advanced wireless communications and next-generation network architectures. In this regard, the crucial role of paradigms such as edge computing and cellular-based sidelink communications is to improve environmental awareness, optimize real-time data management, and ensure reliable, low-latency delivery for safety-critical applications [12]. Additionally, V2X sidelink signals have been presented as a viable solution for vehicle localization, taking advantage of wavefront curvature to estimate the distance and angle of arrival without the need for external infrastructure or strict synchronization, thereby achieving high-precision, low-latency localization in applications such as platooning [13]. Future vehicular networks will also be intrinsically linked to the development of 6G infrastructure, which will prioritize the demands of vehicular applications such as autonomous driving, extended reality, and wireless power transfer [14]. These networks will not only overcome the limitations of their 5G and 4G predecessors but also integrate edge computing for faster decision-making in critical scenarios [15]. Finally, the evolution towards a Social Internet of Vehicles (SIOV) reinforces the collaborative dimension of ITS, where managing social relationships between vehicles using artificial intelligence (AI) and optimizing information flow with software-defined networking (SDN) represent key research areas for the development of socially connected and technologically advanced transportation ecosystems [16]. This article seeks to address these technological trends, providing a comprehensive perspective on the convergence of AI and SDN in VANET and their impact on the future of ITS.

The evolution of VANET into smarter and more connected transportation systems has generated growing interest in exploring new technologies and communication paradigms. The convergence of emerging technologies, such as AI and SDN, offers a promising path to address challenges inherent to VANET, such as managing large volumes of data, security, and privacy. With its machine and deep learning capabilities, AI can revolutionize data management in VANET, enabling more accurate anomaly detection, traffic prediction, and the optimization of network resources. SDN offers greater flexibility and programmability in network management, making

it easier to adapt to dynamic traffic conditions and implement more robust security policies. Innovative solutions can be developed by combining these technologies to improve the efficiency, security, and reliability of VANET, thereby paving the way for smarter and more sustainable transportation systems.

The increasing demand for smart and sustainable mobility solutions has driven the investigation of new technologies to address challenges associated with urban transportation. The convergence of AI and SDN in VANET offers the unique opportunity to develop more efficient, safe, and connected transportation systems. By analyzing how these technologies complement and enhance each other, this study contributes to the existing literature and lays the groundwork for future research in this field. Furthermore, the results obtained may have significant implications for the development of intelligent transportation systems by proposing innovative ideas to improve road safety, reduce traffic congestion, and optimize resource use.

This paper provides an overview of artificial AI techniques used in VANET, including machine learning (ML) techniques, deep learning, and swarm intelligence. In addition, routing challenges in VANET are discussed and SDN is presented as a solution to address these challenges. SDN protocols, controllers, and architectures are described, and the advantages and disadvantages of their application in VANET are analyzed. Table 1 presents a list of the abbreviations used in this review.

As previously mentioned, the evolution of VANET and their integration with technologies, such as AI and SDN, presents a complex and multifaceted panorama that requires systematic analysis. Therefore, a review of scientific literature was conducted to define a search strategy that would allow the identification of the most relevant publications in this field. The methodological process is detailed in the following sections, including the databases consulted, key terms used in the searches, and selection criteria applied.

A. SEARCH STRING

Two bibliographic databases were used to prepare this paper: Clarivate Web of Science (WoS) and Scopus, which were selected because of their compatibility with the ScientoPy tool used to perform automated information preprocessing, such as deduplication and bibliometric analysis. Considering only document-type papers during the period between January 1, 2019, and July 31, 2024, this time range was selected with the aim of capturing the most recent and relevant research, given that VANET, AI, and SDN advances have experienced a significant acceleration in innovations and practical applications in recent years. The search strings used for this analysis were (“Vanet * ” or “Vehicular ad hoc network*”) and ((“machine* learning*” or “ML”) or (“SDN*” or “software defined networking” or “Software-Defined Networking”)). This string was applied to the topic

TABLE 1. List of abbreviations.

Acronym	Definition
ABTS	Adaptive beacon time synchronization
ACO	Ant Colony Optimization
ADMM	Alternating Direction Method of Multipliers
ANN	Artificial Neural Networks
AODV	Ad hoc On-Demand Distance Vector Routing
BSM	Basic Safety Messages
CNN	Convolutional neural networks
CR	Cognitive Radio
CR-VANET	Cognitive radio-vehicular Ad-Hoc Network
C-V2X	Cellular Vehicle to Everything
DDoS	Distributed Denial-of-Service
DoS	Denial-of-Service
DRL	Deep reinforcement learning
DSDV	Destination Sequenced Distance Vector
DSRC	Dedicated Short-Range Communications
DT	Decision Tree
eNB	Evolved Node B
FNN	Feedforward Neural Networks
GPS	Global Positioning System
GRU	Gated Recurrent Unit
HABE	Hierarchy Attribute-Based encryption
HDFS	Hadoop Distributed File System
H-WDFOA	Hybrid Whale Optimization Algorithm
IA	Artificial intelligence
IoT	Internet of Things
IoV	Internet of vehicles
ITS	Intelligent Transportation Systems
kNN	k-Nearest Neighbors
LR	Linear Regression
LSTM	Long Short-Term Memory
LTE	Long Term Evolution
MAC	Medium Access Control
MANET	Mobile Ad hoc Network
MEC	Multi-access Edge Computing
MGM	Maximum Mixing Set
ML	Machine Learning
mmWave	millimeter wave
mPBFT	modified practical byzantine fault tolerance
NB	Naive Bayes
NDN	Named data networking
OBU	On-Board Unit
p-CIDS	Private collaborative intrusion detection system

TABLE 1. (Continued.) List of abbreviations.

PSO	Particle Swarm Optimization
QoS	Quality of Service
RBF	Radial Basis Function
RBFT	Redundant Byzantine Fault Tolerance
RF	Random Forest
RGA	RSU-based Group Authentication
RL	Reinforcement learning
RSU	Road Side Units
SDMEV	Software-Defined Network-based multi-access edge computing framework for the vehicular networks
SDN	Software-defined networking
SDVN	Software-defined vehicular network
SVM	Support vector machine
TDCR	Triple to Double Coincidence Ratio
THz	Terahertz
AAV	Autonomous Aerial Vehicle
V2I	Vehicle to Infrastructure
V2N	Vehicle to Network
V2P	Vehicle to Pedestrian
V2V	Vehicle to Vehicle
V2X	Vehicle to Everything
VANET	Vehicular Ad-Hoc Network
VEC	Vehicular Edge Computing
WAVE	Wireless Access in Vehicular Environments
WiMAX	Worldwide Interoperability for Microwave Access

search in WoS and Scopus, which included title, abstract, and author keywords. Using these search criteria, the dataset was extracted until July, 2024. From the literature review, the main technical challenges and proposed solutions in the field of VANET were identified, with particular focus on the application of AI techniques and SDN.

B. DATA SET

In this study, a Python script called ScientoPy was used to preprocess the information to improve the reliability and accuracy of the data, specifically for the elimination of duplicate documents. To ensure data integrity, the tool identifies duplicate samples by exact matches between the title and authors. In the case of duplicates present in different databases, we decided to retain WoS publications, while eliminating the Scopus sample from the set because of the script design, which prioritizes WoS records. This process resulted in 383 remaining documents, as detailed in Table 2, which shows the number of documents by type and the duplicates eliminated for each database. Fig. 1 shows the total number of documents uploaded and the percentage of documents eliminated.



FIGURE 3. Organization of the survey.

E. MOTIVATIONS

The evolution of ITS is the key to improving the safety, efficiency, and sustainability of urban traffic. In this context, VANET face significant challenges owing to their dynamic nature and increasing demand for real-time data. Efficient management of large volumes of information, ensuring security and privacy, and optimizing vehicular flow requires innovative solutions. This review article analyzes the latest technological trends in VANET, with a special emphasis on AI and SDN as converging approaches to address these challenges.

This review provides insights into how AI improves data management, anomaly detection, and resource optimization in VANET. It also explores how SDNs bring flexibility and programmability to network management, allowing adaptation to dynamic traffic conditions, and establishing more robust security policies. Furthermore, the integration of VANET with emerging technologies, such as blockchain, edge computing, and cognitive radio, is discussed, providing a foundation for advancing the development of smarter and more connected transportation systems.

F. CONTRIBUTIONS

This study focuses on a comprehensive analysis of the crucial roles of AI and SDN in the evolution of VANET. The main objective is to provide new research perspectives by examining how these technologies can improve the management of the growing amount of data, strengthen network security, and optimize vehicular traffic flow in VANET.

1. **AI Analysis in VANET:** This article provides a review of how AI can improve security, traffic efficiency, and information management in VANET.
2. **SDN as a Solution to VANET Challenges:** this article analyzes the potential of SDN to address VANET challenges such as centralized management, flexibility, and scalability.
3. **Synergy between AI and SDN and integration with other technologies:** this paper explores the synergy between

AI and SDN in the context of VANET, examining how AI can enhance SDN capabilities such as intelligent decision making and dynamic adaptation to changing network conditions. In addition, the integration of VANET with other technologies such as edge computing, blockchain, and cloud computing is discussed, highlighting the impact of this convergence.

4. **Challenges in VANET implementation:** this article presents the challenges inherent in the implementation of VANET, including managing large volumes of data in real-time and ensuring the security and privacy of information.

II. VANET REVIEW PAPERS

The literature review on VANET, IoV, and related technologies, such as edge computing, AI, SDN, and blockchain, shows a significant interrelationship between the topics addressed in different studies. These reviews provide a solid foundation for understanding trends, challenges, and opportunities in the field.

The reviewed papers present various approaches and key contributions to these areas. The most significant relationships between the topics addressed are detailed below.

The use of Vehicular Edge Computing (VEC) has been discussed to improve the efficiency and capacity of vehicular networks by bringing computational resources closer to vehicles. This is reflected in reviews highlighting its potential for optimizing ITS [6].

Several studies have explored the application of AI and ML in vehicular networks, highlighting their potential to improve safety, traffic efficiency, and information management in VANET and IoV [5], [7], [11], [17], [18]. ML has proven particularly useful for vehicle classification, traffic management, and safety [2], [19]. In [20], the use of federated learning in IoV was discussed, highlighting its advantages in vehicular perception and improving IoV applications. Although its potential is significant, further research is required to develop suitable frameworks for various applications.

Meanwhile, [21] explored how blockchain technology coupled with AI can improve the security of vehicular networks. Although its implementation presents challenges, the potential of blockchain technology to secure transactions and communication is significant. Papers [2], [22], [23], and [24] discuss the challenges of clustering and routing algorithms in vehicular networks, highlighting the need to improve the stability and efficiency of these mechanisms to adapt to the dynamic nature of VANET. Reference [25] provided a comprehensive overview of the use of cognitive computing and CR-VANET in combination with ML, highlighting the importance of these technologies in optimizing the efficiency and adaptability of VANET.

They also pointed out challenges, such as the need for large datasets and the high computational demand of these algorithms, which could limit their implementation in highly dynamic vehicular network environments [9], [11].

Security remains a critical issue in vehicular networks, especially with the rise of driverless vehicles and ITS. Studies have identified key vulnerabilities, such as GPS spoofing and threats to stereo vision systems, underlining the importance of developing robust authentication and trust management mechanisms [4], [21], [26], [27], [28].

Mobility challenges in vehicular networks, particularly in the context of IoV, are another key focus of this review. Previous studies discussed the importance of developing secure and efficient routing protocols that can adapt to the dynamic characteristics of networks [29], [30], [31]. The integration of next-generation technologies such as 5G mobile communication and edge computing is essential for improving handover management and routing efficiency in these systems [32], [33].

Software-defined networks have been extensively explored for their ability to improve vehicular network management by providing greater flexibility, scalability, and adaptability. This is relevant in VANET, where high mobility and changing topology require dynamic and efficient routing solutions [8], [10], [33], [34], [35]. Despite their advantages, current SDN architectures face limitations, such as single points of failure and controller overload, which has motivated research to develop more robust and decentralized solutions [36].

Table 3 presents a summary of the review articles analyzed in this section, highlighting those that provide a broad overview of VANET and their associated technologies. This table provides the approaches and technologies used and serves as a basis for discussing the interrelations between topics. Fig. 4 classifies the technologies used in review articles.

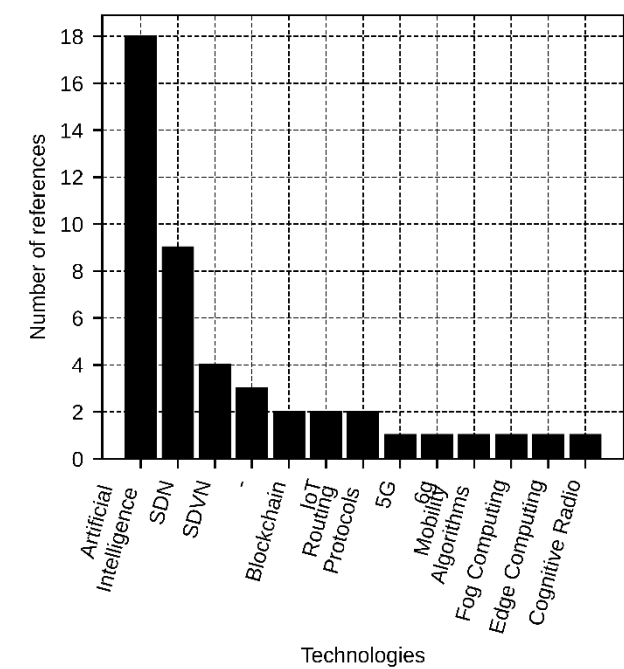


FIGURE 4. Types of technologies found in the search for the papers.

TABLE 3. Review papers on VANET and related technologies.

Year	Reference	Main focus	Technologies used
2019	[7]	Integrating edge computing with VANET	Artificial intelligence
	[17]	Using machine learning in security, communication and traffic in VANET	Artificial intelligence
	[24]	Clustering algorithms in vehicular networks	SDN, Routing Protocols
	[32]	Evolution of VANET towards IoV with IoT	5G, SDN
	[36]	SDN integration in vehicular communications	SDN
	[4]	Safety in driverless vehicles	-
	[8]	Using artificial intelligence to improve V2X systems	SDN
	[20]	Vehicle classification methods in ITS	Artificial intelligence
2020	[22]	Blockchain and machine learning to improve vehicle safety	Artificial Intelligence, Mobility Algorithms
	[25]	Routing Protocols in IoV	Artificial Intelligence, CR
	[28]	Cybersecurity risk management in smart cities with machine learning	SDVN
	[30]	Trust Management in IoV	IoT
	[31]	Using IoT for accident detection and prevention	IoT , Artificial Intelligence
	[35]	Routing in SDVN	SDN, SDVN
	[5]	Using machine learning in security, communication	Artificial intelligence
	[6]	and traffic in VANET	Edge computing
2021	[9]	Improving vehicular communications with SDN	-
	[19]	Artificial intelligence techniques in VANET	Artificial intelligence
	[26]	Machine learning in CR-VANET	-
	[27]	Authentication schemes in VANET	Artificial intelligence

TABLE 3. (Continued.) Review papers on VANET and related technologies.

2022	[33]	Handover management in VANET with 5G	SDVN
	[34]	Architecture and routing in SDVN	SDVN
	[2]	Routing mechanisms in V2X	Artificial intelligence
	[10]	Technologies for data collection and transmission in VANET	SDN
	[11]	Using SDN in VANET	Artificial intelligence
	[18]	Using machine learning in VANET and other fields	Artificial intelligence
	[21]	Application of federated learning in IoV	Blockchain , Artificial Intelligence
	[23]	Clustering algorithms in VANET	Artificial intelligence
	[29]	SDVN Architectures and their Security	Artificial intelligence
	[13]	Architecture for 6G	6G, artificial intelligence
2024	[37]	Vehicular network architecture and security	Artificial intelligence
	[38]	Security challenges	SDN
	[39]	Architecture and routing	SDN, routing protocols
	[This work]	AI and SDN for security, routing, and optimization in VANET	Artificial intelligence, SDN, fog computing, blockchain

III. VANET AND ARTIFICIAL INTELLIGENCE

Artificial intelligence is rapidly transforming in many fields, and VANET research is no exception. AI techniques, particularly ML and deep learning, have proven to be promising for addressing various challenges in VANET systems.

Machine learning, a branch of AI that enables systems to learn from data and make predictions, has been widely applied in VANET research. For example, ML algorithms have been used for intrusion detection [40], traffic prediction [41], and analog beam selection [42]. These algorithms can analyze large amounts of data generated by vehicles and sensors to identify patterns and anomalies, leading to more secure and efficient VANET systems [43]. Furthermore, the potential of ML algorithms has been demonstrated

for detecting spoofing attacks [44], where Basic Security Messages can be analyzed to identify malicious transmissions [45].

VANET are transforming transportation safety and efficiency; artificial intelligence (AI) has played a crucial role in this evolution. As a core component of ITS, VANET enable vehicles to communicate with each other and with the infrastructure to share essential information regarding road safety and traffic conditions [46], [47], [48]. However, challenges, such as data congestion, safety, and the need for real-time responses, require advanced solutions. AI, with its ML and predictive analytics capabilities, has an exceptional potential. Various AI techniques, such as deep learning, neuro-fuzzy algorithms, and support vector machines (SVM), have been integrated into VANET architectures to address these challenges [47], [49], [50].

ML algorithms such as SVM, KNN, RF, Boosting, and Ensemble Voting are highly effective in improving the security and reliability of communication in VANET [51]. By analyzing the communication patterns and vehicle behavior, ML algorithms can detect anomalous behaviors that could indicate a security threat. However, the similarity between normal and abnormal data can pose challenges to ML algorithms, and the possibility of attackers fooling ML models by introducing adversarial data remains an open research topic [51].

Reinforcement learning (RL), an ML paradigm that enables agents to learn through interactions with their environment, is gaining increasing interest in VANET research [52]. RL algorithms can be used to optimize medium access control (MAC) protocols [53], routing schemes [54], and resource scheduling [55] in VANET. In particular, deep reinforcement learning combining RL with deep neural networks has been shown to be effective in handling complex dynamics and uncertainty in VANET systems [56]. For example, deep reinforcement learning can be used to make intelligent task offloading decisions for VANET in fog computing environments by considering the resource availability and estimated vehicle locations [24].

AI enables VANET to analyze large volumes of data generated by sensors and V2V/V2I communications, thereby facilitating the detection of anomalies, prediction of potential risks, and optimization of traffic flow. For example, ML algorithms can identify patterns in sensor data to detect dangerous driving behaviors or road conditions, alert drivers, and prevent accidents [57], [58].

Despite the advances in AI for VANET, several challenges and research gaps remain. One limitation is the need for large-scale and realistic datasets to train and evaluate the AI algorithms. Most existing datasets are relatively small or lack the diversity needed to capture the complexity of real-world scenarios [40]. Furthermore, the development of robust and attack-resistant AI algorithms remains a challenge. Attackers may attempt to trick or manipulate AI algorithms, highlighting the need for robust defence mechanisms [59], [60].

TABLE 4. Matrix of AI Algorithms for VANET by category and usage. The symbol ✓ indicates that the category was used and the symbol • indicates that the category was not used.

Reference	Algorithms	Classification	Optimization	Inference	Prediction	Regression	decision making	Clustering	Optimal control	Using AI algorithm	Aspect of VANET that improves
[40]	LR, NB, DT, SVM, kNN, RF, AdaBoost, XGBoost	✓	•	•	•	•	•	•	•	Binary and multiclass classification for intrusion detection.	Security
[41]	LSTM, GRU, RF	•	•	•	✓	•	•	•	•	Network traffic prediction considering road traffic parameters.	Detecting a variety of attacks.
[42]	SVM	✓	•	•	•	✓	•	•	•	Analog beam selection to improve average sum rate in concurrent transmissions.	Network efficiency
[43]	SVM, GA, ACO, PSO	•	•	•	•	•	•	•	✓	SVM optimization for intrusion detection using genetic algorithms, ant colony optimization, and particle swarm optimization.	Network traffic prediction and management.
[44]	kNN, RF	✓	•	•	•	•	•	•	•	Position spoofing detection using information from two consecutive BSs.	Network efficiency
[45]	kNN, RF, Ensemble Learning	✓	•	•	•	•	•	•	•	Position spoofing detection using a public dataset (VeReMi) and ensemble learning.	Improves analog beam selection in mmWave communications.
[46]	Logic diffuse	•	•	✓	•	•	✓	•	•	Real-time driving support system for risk management.	Security
[47]	SVM, H-WDFOA	✓	✓	•	•	•	•	•	•	Improving communication robustness in VANET using a hybrid machine learning model.	Intrusion detection .
[48]	PSO	•	✓	•	•	•	•	•	•	Clustering- based routing using PSO to select cluster heads.	Security
[49]	Neuro-fuzzy	•	•	✓	•	•	✓	•	•	Privacy-preserving authentication scheme in fog-assisted VANET.	Detection of vehicles that falsify their position.
[50]	LSTM	✓	•	•	•	•	•	•	•	Detecting fake messages using an LSTM neural network for time series classification.	Security
[51]	SVM, kNN, NB, RF, Ensemble Learning, Voting Ensemble Learning	✓	•	•	•	•	•	•	•	Classification of vehicle behavior to detect anomalies.	Detection of vehicles that falsify their position.
[52]	RL	•	•	•	•	•	✓	•	✓	Route selection in VANET .	Security
[53]	Q-Learning	•	•	•	•	•	•	•	✓	MAC in IEEE 802.11p vehicular networks.	Reducing the risk of accidents.
[54]	ANN	✓	•	•	•	•	•	•	•	Predict vehicle arrival rate in real time.	Network performance
[55]	RL	•	•	•	•	•	•	•	✓	Cluster-based cooperative programming for high-mobility vehicular networks.	Improves the robustness of communication.
[56]	Deep Q-network	•	•	•	•	•	•	•	✓	Fog node offloading strategy based on Deep Q- network for 5G VANET networks.	Network performance

TABLE 4. (Continued.) Matrix of AI Algorithms for VANET by category and usage. The symbol ✓ indicates that the category was used and the symbol • indicates that the category was not used.

[57]	RF, K-means	✓	•	•	•	•	•	✓	•	Congestion control in VANET using blockchain, dynamic service function chaining and machine learning.	Route optimization.
[58]	FNN	✓	•	•	✓	•	•	•	•	Onboard prediction that captures anomalous vehicle responses and performs mitigation in real time.	Security
[59]	Fuzzy logic guided genetic algorithm	•	✓	•	•	•	•	•	•	Resource optimization in 5G VANET networks, including bandwidth allocation.	Authentication and privacy preservation.
[60]	Dempster-Shafer theory, Bayesian classifier	✓	•	•	•	•	•	•	•	Intrusion detection in intelligent vehicular networks.	Security
[61]	Regression Logistics, NB	✓	•	•	•	•	•	•	•	Creation of a global classifier through collaborating neighboring vehicles.	Detection of fake messages.
[62]	SVM, Logistic Regression	✓	•	•	•	•	•	•	•	Secure data exchange mechanism coupled with a cyber-attack detection approach for intelligent vehicles.	Security
[63]	ANN, SVM	✓	•	•	✓	•	•	•	•	Predicting mobility.	Detection of different types of vehicle misbehavior.

The dynamic and distributed nature of VANET pose additional challenges for AI applications. AI algorithms must be able to adapt to rapid changes in network topology and vehicle behavior. Furthermore, they must be computationally efficient to satisfy the resource constraints of the vehicles and roadside units. Future studies should address these limitations by developing richer datasets, robust, and efficient AI algorithms, and distributed architectures for AI processing in VANET.

Table 4 presents a classification of 25 articles from the categories of research and development or implementation on VANET and how AI is used to improve various aspects of this technology, identifying the AI algorithms used, their specific application, and the aspects of the VANET that are sought to be improved.

Table 5 categorizes the selection of papers on VANET. The papers were classified according to their focus, including intrusion detection, resource management, access control, mobility prediction, and security and privacy.

The inherent dynamics of VANET pose significant challenges in AI applications. Constant vehicle mobility, varying density, and ever-changing topology make it difficult to create accurate prediction models and efficiently manage resources [41], [42], [47], [63]. The changing nature of the network also complicates training and updating detection models, such as intrusion detection, because the relevant features can change rapidly [45].

Security and privacy are paramount concerns for VANET. Accurate detection of attacks such as DoS and DDoS, while minimizing false alarms, is critical [51]. Furthermore, the inference of sensitive information from the data used to train

AI models poses significant risks [61]. Therefore, balancing the accuracy of a model using privacy protection is a key challenge. Moreover, the selection of relevant features and an appropriate learning algorithm are crucial for the success of intrusion detection systems [42], [45], [62].

Additionally, the heterogeneity of the devices and components inherent in VANET complicates the implementation of uniform AI solutions. Variability in the computational and communication capabilities of vehicles makes it difficult to deploy complex algorithms and process large amounts of data in real-time. Moreover, the scalability of AI systems as the number of vehicles in the network increases is a major challenge, because efficient solutions are required in terms of both computation and storage [41], [42].

Fig. 5 shows the distribution of papers on VANET networks classified according to their main focus. Security is the most researched area, followed by routing and congestion control.

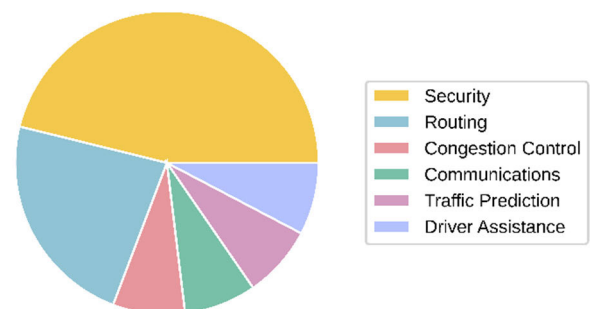


FIGURE 5. Papers by application area in VANET.

TABLE 5. Papers by area of application.

Area of application	Papers
Security	[40], [43], [44], [45], [47], [49], [50], [51], [58], [60], [61], [62]
Routing	[48], [52], [54], [55], [56], [59]
Congestion control	[53], [57]
Communications	[42], [57]
Traffic prediction	[41], [63]
Driver Assistance	[46], [58]

Despite the growing interest in applying AI to VANET, the current literature presents gaps that require further analysis. A common assumption is the availability of large-scale and realistic datasets for training and evaluating AI algorithms [40]. However, most datasets are limited in size or diversity, which makes it difficult to generalize the models to real-world scenarios [40]. This limitation can affect the accuracy and effectiveness of the AI solutions in practical applications. Moreover, the selection of relevant features and an appropriate learning algorithm remain areas of uncertainty [42], [45], [62]. Although various techniques have been applied, from traditional machine-learning algorithms to advanced deep-learning approaches, the lack of justification for choosing specific methods for VANET scenarios makes it difficult to benchmark current proposals [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63]. A thorough analysis is required to assess not only the performance and accuracy of the models, but also their adaptability to VANET dynamics, computational efficiency, and resilience to adversarial attacks [51], [52], [59], [60]. Furthermore, although the importance of security and privacy has been recognized, these areas have received less attention than the model accuracy [42], [45], [51], [61], [62]. The inference of sensitive information from training data coupled with the lack of robust mechanisms to ensure user privacy highlights the need for innovative approaches that balance efficiency with information protection in AI solutions for VANET.

IV. VANET AND SOFTWARE-DEFINED NETWORKING

SDN have revolutionized the management and optimization of VANET, which are essential for the development of more secure, efficient, and reliable ITS [1], [3], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75].

One of the main benefits of SDN in VANET is its ability to provide centralized network management through an SDN controller [3], [67], [70], [71], [72], [73], [75], [76], [77], [78]. This controller has a global view of the network and can optimize system performance in real time, adapting to

changing traffic conditions and application requirements [3], [68], [70], [76], [78]. The separation of the control plane from the data plane allows for greater flexibility and programmability, facilitating the implementation of security policies, resource management, and routing optimization [3], [64], [70], [71], [72], [77], [79], [80], [81], [82], [83].

Security is another critical aspect of VANET, where the open and wireless nature of communication renders them vulnerable to various attacks [64], [70], [80]. SDN can improve security by providing centralized control of security policies and enabling faster response to threats [70], [71], [77], [78]. Various sources have proposed the use of authentication techniques, such as RSU-based group authentication and intrusion detection systems, to detect and mitigate potential attacks [70], [80], [84].

Incorporating technologies such as fog computing along with SDN can further improve the performance and efficiency of VANET [1], [71], [72], [73], [79]. Fog computing enables data processing and decision-making closer to data-generating devices such as vehicles, thereby reducing latency and freeing up network resources [1], [71], [72], [73]. The combination of SDN and fog computing facilitates the deployment of latency-sensitive services such as collision warnings, driving assistance, and real-time traffic information sharing [71], [72], [73], [79].

Research in the field of SDN for VANET focuses on developing efficient routing algorithms and protocols that take full advantage of the benefits of this architecture [1], [68], [72], [76], [79], [81], [85], [86], [87]. Approaches such as Quality of Service (QoS)-based routing, multipath routing, and failover routing have been explored to improve the reliability, latency, and resource utilization in VANET [1], [65], [68], [72], [76], [79], [81], [86]. The integration of ML and blockchain techniques is a promising approach to optimize network performance and enhance security and trust in SDN-based VANET [1], [3], [65], [70], [80], [78], [85], [86], [88].

Future research on the SDVN could focus on addressing research gaps and exploring new avenues. This includes the development of more sophisticated resource-management algorithms, privacy-preserving techniques, and secure and resilient SDVN architectures. This evolution of SDVN has the potential to radically transform ITS, making road transport safer, more efficient, and smarter, as SDN has potential as a key technology to address VANET challenges and enable the next generation of ITS. Centralized network management, enhanced security, integration with fog computing, and the development of advanced routing protocols are active research areas that are paving the way for safer, more efficient, and more connected transportation.

Table 6 presents the group of articles from the research and development or implementation categories related to VANET, focusing on the application of technologies such as SDN.

Fig. 6 shows the distribution of VANET-related studies using SDN technologies. It can be observed that the most

researched areas are network management, routing, security, and connectivity. These categories cover a broad spectrum of challenges in the VANET field, from traffic optimization to data privacy protection.

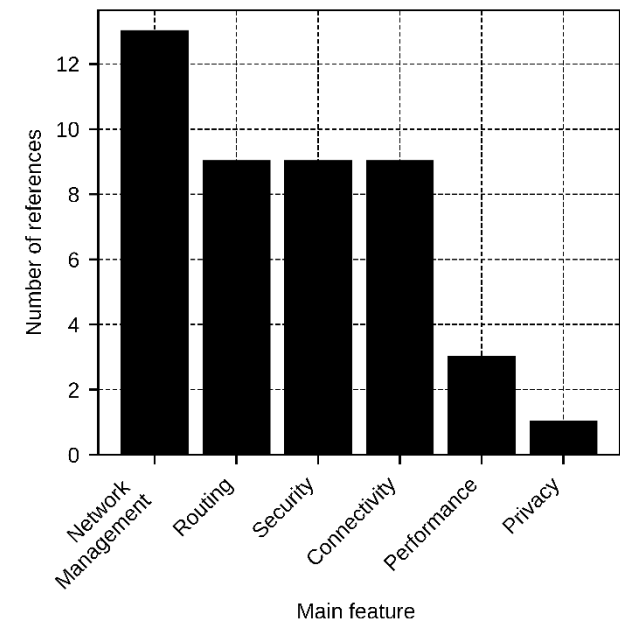


FIGURE 6. VANET research with SDN by category.

Table 7 provides an overview of various SDN architectures applied in VANET along with the specific technologies used in each case.

SDN integration in VANET represents a significant step towards autonomous driving and intelligent traffic management. However, this convergence poses several challenges that must be addressed to ensure the security, efficiency, and scalability of these networks. Consulted studies [65], [67], [71], [82] have shown that one of the main obstacles in the implementation of SDN in VANET is unstable connectivity. The high mobility of vehicles causes constant fluctuations in the network topology, which makes it difficult to maintain reliable communication with a centralized SDN controller. This situation directly affects the efficiency of the network, especially in critical applications, such as autonomous driving, which require minimal latencies and high availability.

Another key challenge is the security. The centralization inherent in SDN exposes the network to various threats, such as denial of service (DoS) attacks and data manipulation [3], [64], [65], [68], [78], [86]. Furthermore, protecting user privacy is critical because large amounts of data are collected from vehicles [3], [75], [76], [77], [81], [82].

Security in VANET poses significant challenges because of their dynamic and distributed nature. Although the centralization of SDN facilitates its management, it introduces significant vulnerabilities. Common attacks include denial of service, which can disrupt critical communication by overloading the network and data manipulation,

where attackers can falsify traffic or crash information, thereby causing chaos. Sybil attacks allow malicious nodes to assume multiple identities and degrade information in the network [90].

Middle attacks also stand out, where attackers intercept and modify messages between vehicles or infrastructure, thereby affecting the confidentiality of the information [91]. User privacy is another key concern because the collection of sensitive data, such as location and speed, can be tracked if adequate anonymization and identity protection mechanisms are not implemented [90], [92].

To mitigate such risks, it is crucial to adopt robust security strategies. This includes the strong authentication of nodes to prevent spoofing attacks [91], the use of cryptography to ensure data integrity [90], and the implementation of intrusion detection mechanisms that monitor the network for suspicious activity [91]. Furthermore, privacy should be integrated into the design of VANET by employing anonymization and secure computing techniques [90], [91].

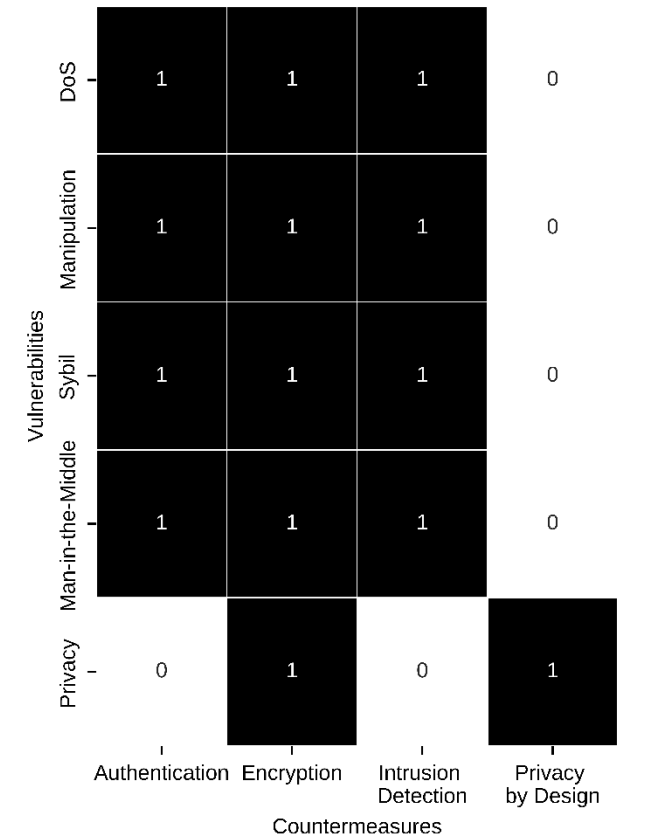


FIGURE 7. Matrix of attack areas and countermeasures for VANET networks.

Fig. 7 presents a matrix of the attack areas and countermeasures in the VANET. Each row represents an attack type and each column corresponds to a security technique. A value

TABLE 6. State of the art in VANET research with SDN.

Reference	Aspect that improves	Main feature	Aspects to improve
[1]	Routing performance	Hierarchical SDN architecture and fog computing	Detail IDVR protocol and use of SDN controller
[3]	Real-time urban traffic analysis	Traffic congestion detection and prediction	Improve performance with external parameters (vehicle density)
[64]	Data processing efficiency in large-scale VANET.	SDN scalable and adaptable controllers	Trust modeling on blockchain, vehicle transition probabilities
[65]	Network management and security in ITS.	Separation of control and data planes	Calibration and validation for urban planning, vehicle network management, emergency systems, etc.
[66]	Access control in the user interface	BENBI scheme for secure access control	Deeper evaluation of performance in real-world scenarios
[67]	Network selection and data dissemination	Communication model for various applications	Addressing data dissemination challenges in SDVN (congestion management, QoS)
[68]	Intrusion Detection	Hybrid Machine Learning Model	Consider the impact of adversarial attacks and explore defense strategies
[69]	Vehicle route planning	SDN architecture for monitoring and emergency response	Develop robust replanning strategies for accidents
[70]	Computing and routing latency.	SDMEV integrating eNB with RSU	Consider using device-to-device LTE to disseminate V2X messages
[71]	Privacy of communications	Combining SDN and Fog Computing for Privacy	Reduce bandwidth consumption at fog computing level
[72]	Data transmission	Route stability estimation, fog nodes, SDN controller	Develop task assignment technique based on mobility and speed
[73]	Proactive load balancing in IoV networks	Vehicles parked as fog computing nodes	Consider resource costs associated with the use of vehicles
[74]	Mitigation of traffic congestion	Traffic management with SDN and heuristic algorithms	Incorporating machine learning for route optimization and congestion prediction
[75]	Network performance	System management through programmed software	Security (virtual attacks, internal threats, secure routing, policy lifecycle)
[76]	Routing in SDVN.	Optimization-based routing algorithm	Inclusion of future connections, precise modeling of transmission range
[77]	Trust Management System	Trust management system with blockchain	Resistance to coordinated attacks against multiple RSUs
[78]	Transmission latency and data dissemination reliability.	MEC server on base station for caching	Consider impact of different network environments (urban vs. rural)
[79]	Data dissemination efficiency	Fog computing architecture	Study of security and privacy aspects in data dissemination
[80]	Energy-efficient end-to-end security	SDVN framework with RGA and p-CIDS	Comparison with other security mechanisms, evaluation of the impact of network parameters
[81]	Data collection in DSRC and C-V2X hybrid vehicular networks.	TDCR mechanism for QoS and cost reduction	Integrating data aggregation strategies within the cluster

TABLE 6. (Continued.) State of the art in VANET research with SDN.

[82]	Routing	SDN architecture to improve routing	Incorporate privacy mechanisms to protect sensitive information
[83]	Reliability and security of data dissemination.	SDN-based network coding for secure broadcast	Explore multidimensional flow extent for all outcomes
[84]	DDoS	Intrusion detection system based on RBF-SVM	Use more comprehensive and representative data sets for DDoS attack assessment
[85]	Efficiency in data transmission in vehicular environments	Road segmentation and spectrum detection with machine learning	Consider more critical CR network parameters and dynamic threshold values
[86]	Connectivity and routing	Hybrid routing architecture based on link stability	Large-scale performance evaluations with realistic traffic models
[87]	Efficiency in the flow of information and response times	SDN-assisted technique in heterogeneous vehicle organizations	Detailed analysis of failover and recovery mechanisms for SDN controller
[88]	Security and trust in IoV networks	Blockchain, SDN, fog computing, reputation-based trust model	A more in-depth analysis of alternative consensus protocols to mPBFT
[89]	Security and efficiency in resource allocation	Decentralized trust management with blockchain	Addressing privacy challenges in data storage and sharing on blockchain

of “1” indicates that the countermeasure is applicable to mitigate the vulnerability, while a “0” indicates that it is not.

Therefore, efficient resource management is a challenge. The dynamic allocation of bandwidth and transmission power in a dynamic and heterogeneous environment, such as a VANET, requires sophisticated algorithms and adaptive control mechanisms. Furthermore, scalability is essential for managing the increasing number of vehicles and volume of data traffic [1], [66], [67], [77], [82], [83], [84], [85], [87], [89].

SDN implementation in VANET is an active and promising research field although it faces several significant challenges. VANET are characterized by their changing topology, high node mobility, and strict latency requirements for time-sensitive applications, such as security and traffic management [72], [74], [86]. Traditional networking approaches often struggle to manage these challenges, resulting in congestion, delays, and a poor service quality. SDN introduces a centralized paradigm by separating the control plane from the data plane [66], [71], enabling flexible and programmable management that optimizes the real-time performance [71], [72]. Furthermore, SDN capabilities allow dynamic configuration of packet forwarding paths based on changing traffic conditions, improving routing efficiency, and reducing congestion [1], [72], [74]. Some promising areas of research include distributed SDN architectures to improve network resilience, blockchain-based authentication mechanisms to ensure transaction integrity and transparency, and the use of AI to develop ML algorithms that are capable of adapting to network conditions. Protocol standardization is also

key to facilitating the interoperability between systems and manufacturers, allowing VANET to overcome their inherent limitations and provide more reliable and efficient communication.

Fig. 8 presents the distribution of the different types of SDN architectures used in the VANET-related research. A predominance of centralized and hierarchical architectures was observed. This suggests that these topologies are the most common and have been explored in current research. However, there is also growing interest in distributed and hybrid architectures that offer greater scalability and flexibility.

Furthermore, the implementation of SDN in VANET should consider the design of inter-vehicular collaboration mechanisms and infrastructure to maximize network efficiency. The use of advanced communication technologies, such as 5G and future 6G, in conjunction with SDN can improve network responsiveness, reduce latency, and enable more robust vehicular communication [1], [3], [65], [71], [78], [86]. These advanced networks also facilitate integration with smart city ecosystems, where real-time data exchange between vehicles and infrastructure can optimize traffic management, reduce accidents, and improve the user experience [65], [70], [71], [80], [84].

Finally, sustainability is emerging as a key objective in the evolution of SDN in VANET. Current research addresses strategies to minimize the energy consumption in vehicular networks, leveraging efficient computing techniques, and adaptive resource usage planning [71], [72], [79], [82]. In this context, the combination of SDN with green computing

TABLE 7. Comparison of SDN architectures for VANET.

Reference	SDN Architecture Type	Technologies used
[1]	Hybrid	Fog Computing, Clustering, OpenFlow, AODV
[3]	Centralized	K-means clustering, LSTM, DSRC and WAVE protocols.
[64]	Distributed	Blockchain, Deep Deterministic Q-Learning, RBFT, CNN.
[65]	Hierarchical	5G, NDN, HBase, HDFS.
[66]	Centralized	Identity-based broadcast encryption, MAC
[67]	Centralized	DSRC, 3G, Wi-Fi, WiMAX.
[68]	Not specific	RF, Coresets, Bayesian ML, SVM, CNN.
[69]	Hierarchical	UAV, 5G,
[70]	Distributed	Multi-access Edge Computing, eNB-type RSUs, interface LTE logic, IEEE 802.11p, LTE device-to-device
[71]	Centralized	Fog Computing, HABE encryption, IEEE 802.11p, LTE 4G, 5G
[72]	Hierarchical	Fog Computing, RSUs, base station
[73]	Hierarchical	Fog Computing, IoV, OpenFlow protocol
[74]	Centralized	DSRC, LTE, Heuristic algorithms
[75]	Centralized and distributed	Elliptic curve cryptography, hash functions.
[76]	Distributed	LTE, DSRC, BSM
[77]	Centralized	Blockchain, 5G, trust management, real-time video reporting.
[78]	Hierarchical	Mobile Edge Computing, IEEE 802.11p/LTE
[79]	Hierarchical	Fog Computing, Network Coding, DSRC
[80]	Distributed	RGA, p-CIDS, ADMM
[81]	Hybrid	DSRC, C-V2X
[82]	Distributed	OpenFlow, Open vSwitch, blockchain, WAVE, LoRaWAN, LTE.
[83]	Centralized	Network Coding, MGM, DSRC, WAVE
[84]	Not specific	Radial Basis Function - Support Vector Machine, Grid Search Cross-Validation, Principal Component Analysis
[85]	Not specific	CR, CR-VANET, segment-based CR-VANET, Fuzzy Naive Bayes algorithm, triagent reinforcement learning algorithm, Energy Detection, Matched Filter Detection
[86]	Hierarchical	DSRC
[87]	Not specific	RSU, I2V
[88]	Not specific	Blockchain, Fog Computing, 5G, IoV
[89]	Centralized	Blockchain

supported by AI technologies can significantly contribute to the creation of energy-sustainable solutions, enabling greener transportation aligned with global sustainability goals [65], [71], [80], [85], [86].

V. VANET INTEGRATED WITH SOFTWARE-DEFINED NETWORKING AND ARTIFICIAL INTELLIGENCE

Artificial intelligence further enhances SDN capabilities in VANET by enabling intelligent decision-making and dynamic adaptation to changing conditions [34], [93], [94].

AI techniques such as fuzzy logic and RL can be used to optimize various aspects of network operations, including data processing layer selection [95], [96], traffic signaling scheduling [93], and routing [95]. For example, in SDN-integrated VANET systems, a fuzzy system can determine the optimal processing layer (cloud, fog, or edge) for application data based on factors such as the data size, time sensitivity, and relative vehicle speed [96], [97]. RL enables the system to adapt to the dynamics of the network topology by adjusting the routing policies during packet transmission [95].

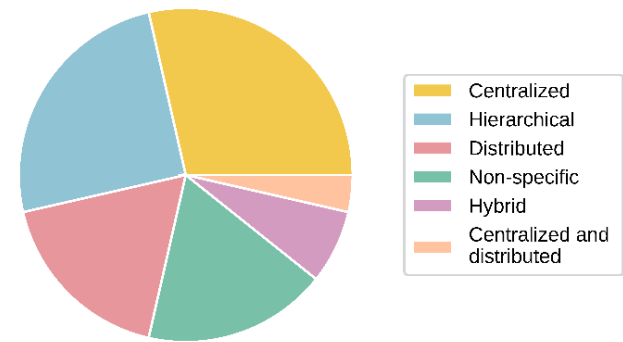


FIGURE 8. Trends in SDN architectures for VANET.

A key aspect of VANET operation is the use of beacon signals. These are periodic messages that vehicles transmit to nearby vehicles in order to announce their presence and status. These signals allow a local map of the network topology to be built, thereby facilitating the maintenance and communication within the network. The information in a beacon signal can include the vehicle’s position, speed, direction, and data on traffic conditions or road status. This mechanism is essential for ensuring efficient and robust connectivity in dynamic environments, such as VANET systems.

The integration of SDN and AI in VANET has the potential to revolutionize transportation systems by enabling smarter, more efficient, and safer traffic management [93], [94]. SDN provides a flexible and programmable architecture for network control [96], [97], [98], whereas AI enables automation and intelligent decision-making to optimize network performance [93], [94], [95]. This powerful synergy can lead to the development of a more robust and adaptable ITS that can cope with the growing demand for connected traffic environments.

Table 8 provides an overview of the research analyzed, grouping them according to the aspect of the VANET they address and the types of AI algorithms and SDN architectures used.

TABLE 8. AI and SDN-based solutions for VANET.

Reference	Aspect that improves	AI algorithms	SDN Architecture Type
[93]	Traffic sign optimization	Decision trees diffuse	Hybrid
[94]	Trust management	Deep Q learning	Centralized
[95]	Routing performance	Fuzzy logic, reinforcement learning	Hierarchical
[96]	Resource management	Logic diffuse	Hybrid
[97]	Resource management	Logic diffuse	Hybrid
[98]	Intrusion Detection	Generative networks antagonistic	Distributed

Fig. 9 shows the AI algorithms used in recent VANET research in combination with different types of SDN architectures. It can be observed that hybrid and distributed architectures tend to be preferred in combination with fuzzy logic, indicating a leaning towards flexible and adaptive approaches to managing connected traffic environments.

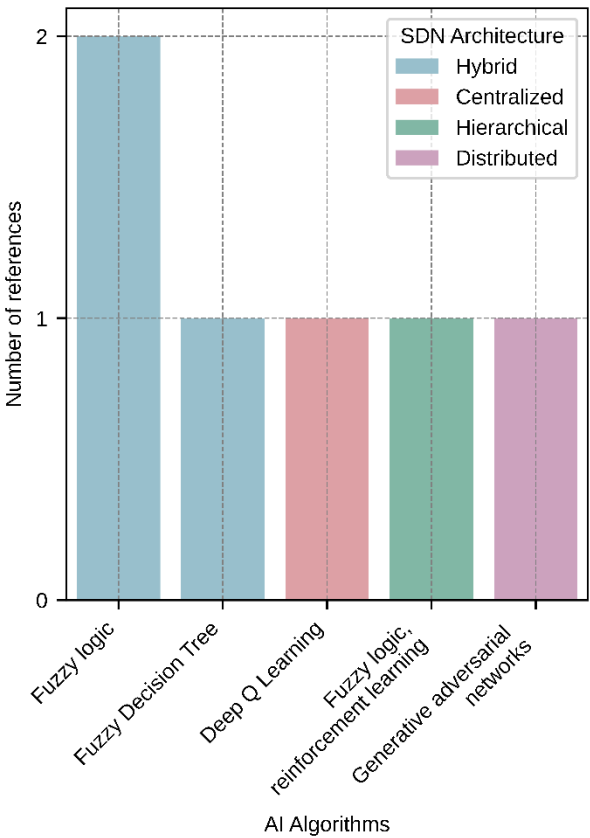


FIGURE 9. AI algorithms by type of SDN architecture.

Routing is a fundamental pillar in the design and operation of VANET not only because of its direct impact on network performance but also because of its unique technical challenges and interdependence with emerging technologies. First, network quality in high-mobility environments critically depends on an efficient routing scheme, as summarized in Tables 9 and 10. This reduces latency, ensures timely packet delivery, and optimizes connectivity, which are essential factors for applications, such as road safety and real-time traffic management. In VANET, where nodes (vehicles and devices) constantly enter and leave the coverage range, optimal route selection is an indispensable requirement for avoiding communication interruptions. Failure in this aspect could lead to the loss of critical data or delays, with potentially serious consequences.

However, routing in VANET presents an inherent complexity that distinguishes them from static or low-mobility networks. Network topology changes dynamically owing

TABLE 9. Details routing protocols/solutions in VANET based on AI and SDN. The symbols “SP”, “MP”, “U”, “MC”, “B”, “L”, “M”, “H”, “D”, “S”, “-” and “X” indicate single path, Multipath, Unicast, Multicast, Broadcast, Low, Medium, High, Dense, Sparse, Include, Un-mentioned and Un-included respectively.

Reference	Protocol / Solution	Year	Data transmission route	Network Topology Maintenance	Transmission mode	Communication model	Routing algorithm	Network Bills general	Message priority	Complexity	Scalability	Reliability	Application data	
[1]	ICDRP-F-SDVN	2021	-	Beacon	B	V2V/V2I	AODY	L	-	-	H	H	-	
[54]	CRS-MP	2018	MP	Predication	OR	V2V/V2I	Bipartite Matching, Graph-Based	L	-	✗	H	M	H	-
[65]	-	2018	MP	Beacon/Predication	U/B	V2V/V2I	Traffic engineering	L	D	✓	H	M	H	Safety, Infotainment
[67]	-	2019	MP	Predication	U/B/ MC	V2V/V2I	-	L	D , S	✓	M	M	M	Safety, infotainment
[70]	SDMEV	2020	MP	Beacon	U/B , MC	V2V/V2I	-	-	-	✓	M	M	M	Safety
[72]	SFSR	2020	MP	Beacon	OR	V2V/V2I	Greedy, Graph-based	L	D , S	-	-	H	-	Safety, infotainment
[76]	-	2019	MP	Beacon/Predication	OR	V2V/V2I	Graph-based	L	-	-	-	-	-	-
[78]	-	2020	-	-	-	V2V/V2I	AODY, AODV	-	D	-	-	H	-	-
[81]	TDCR	2020	-	Predication	MC	V2V/V2I	Graph-based (Dijkstra)	-	-	✓	-	-	-	Safety, Infotainment
[82]	SURFER-1	2020	MP	Beacon	U/B	V2V/V2I	Geographic forwarding, carry-and-forward	L	-	-	M	M	M	-
[83]	SENCoSDD	2019	MP	-	B	V2V/V2I	-	M	-	-	M	M	H	Safety, infotainment
[86]	-	2019	SP/MP	Beacon/Predication	U/B/ MC	V2V/V2I	Graph-based (Dijkstra)	-	D , S	-	-	-	-	-
[87]	-	2022	SP/MP	Beacon	U/B/ MC	V2V/V2I	Graph-based	-	D , S	-	L	-	-	-
[93]	-	2020	-	-	-	V2V/V2I	AODV, DSDV	-	-	✓	-	H	H	Safety, Infotainment
[94]	SD-TDQL	2020	-	-	-	V2V	-	-	-	-	-	-	-	-
[95]	-	2021	MP	-	-	V2V	Greedy	L	-	-	M	M	H	-
[99]	-	2019	MP	-	U/B/ MC	V2V/V2I	Latency-aware offloading algorithm	L	D , S	✓	M	M	M	Safety, infotainment
[100]	-	2020	-	-	-	V2V/V2I	-	-	-	-	-	H	-	Safety, infotainment
[101]	SDN-controlled CR-V2X routing	2020	MP	Beacon/Predication	U/MC	V2V/V2I	Graph-based, Greedy	L	D	✓	H	H	H	Safety, infotainment

TABLE 10. Objectives, innovations and types of AI/SDN in routing protocols/solutions in VANET.

Reference	Protocol / Solution	Performance aim	Innovative idea
[1]	ICDRP-F-SDVN	Improve packet delivery rates, provide more reliable paths in the routing process, and reduce control overhead	The protocol uses a novel hierarchical architecture that combines SDN with fog computing, clustering algorithm, control overhead reduction approach, and dual-phase technique.
[54]	CRS-MP	Minimizing transmission delay in VANET	Using an artificial neural network to predict vehicle movement to improve route planning
[65]	No specific name is mentioned	Improve connectivity between vehicles and SDN controllers, provide high data rates and bandwidth, and improve the overall network efficiency in ITS	Integrating SDN and 5G technology into ITS for flexible and efficient network management with higher bandwidth.
[67]	No specific name is mentioned	Improve network performance compared to existing schemes	Centralized network selection using utility functions and Stackelberg game theory
[70]	SDMEV	Improve latency and reliability of data dissemination in vehicular networks for V2X services.	SDN and MEC were integrated with an eNB co-located with RSUs to create a distributed vehicular network architecture.
[72]	SFSR	Improve data transmission in VANET by switching between transmission over the VANET infrastructure and transmission over the internet when necessary.	SDN and fog computing are used to estimate route stability and dynamically switch between V2V communication and the internet to improve data transmission.
[76]	No specific name is mentioned	Finding multiple routes with lower travel times that are sufficiently stable to deliver a given number of packets.	Routing individual packets on the shortest possible paths that are sufficiently stable in their existence
[78]	No specific name is mentioned	Reduce packet loss and delay transmission during handovers in hybrid vehicular networks.	A two-tier SDN architecture and MEC are used to proactively manage handovers and data caching.
[81]	TDCR	Achieving a balance between end-to-end delay and cellular bandwidth costs in vehicular data collection.	A single-hop clustering approach that considers data transmission between CHs
[82]	SURFER-1	Efficient and secure routing in IoV , especially in scenarios with low node density.	VANET routing with SDN in an RSU network to improve performance in low-density scenarios.
[83]	SENCoSDD	Maximize packet delivery rate, reduce packet loss, and maximize bandwidth efficiency while providing data security.	MGM-based network coding is combined with SDN to improve the reliability and security of data transmission in VANET.
[86]	No specific name is mentioned	Maximize packet delivery by satisfying latency-based QoS.	The framework uses a combination of centralized and distributed routing techniques, including unicast, broadcast, and store-and-forward routing.
[87]	No specific name is mentioned	Improve the efficiency of information transmission and traffic management in VANET, especially in congested situations.	SDN is used to gain a global view of network traffic and intelligently route traffic through proxy nodes, thereby improving response time and performance.
[93]	No specific name is mentioned	Optimize traffic signals to improve traffic flow at urban intersections.	Using a 3-stage fuzzy decision tree model for traffic signal optimization.
[94]	SD-TDQL	Improve the quality of communication links in VANET, particularly under the influence of malicious nodes.	The innovative idea is to combine a trust model to evaluate the behavior of neighboring vehicles and detect malicious nodes.

TABLE 10. (Continued.) Objectives, innovations and types of AI/SDN in routing protocols/solutions in VANET.

[95]	No specific name is mentioned	Improve the stability and efficiency of routing in SDVN, particularly in urban scenarios with dynamic traffic.	Combining fuzzy logic and reinforcement learning for hierarchical routing decision making in SDVN.
[99]	No specific name is mentioned	Reduce latency in V2V and V2I communication.	A multilevel MEC structure is implemented in the VANET architecture to bring data processing closer to vehicles.
[100]	No specific name is mentioned	Minimize latency, energy consumption and flow costs for all vehicles.	Blockchain for secure access control and deep reinforcement learning (DRL) for offloading optimization.
[101]	SDN-controlled CR-V2X routing	Ultra-high data rate and minimized transmission delay.	An SDN-controlled intelligent V2X routing protocol uses predictive switching between the mmWave and THz links.

to vehicle speed, variability in traffic density, and fluctuations in channel congestion. These factors require algorithms that can adapt to real-time conditions and simultaneously evaluate multiple variables. Solving this problem cannot be limited to traditional approaches; it requires hybrid solutions that integrate fast heuristics for immediate decisions and machine-learning models to anticipate scenarios and optimize long-term routes. Finally, the relevance of routing is amplified when considering its interaction with technologies, such as SDN and AI. SDN offers a centralized framework for dynamically managing routing, allowing controllers to adjust routes based on the overall state of the network and traffic conditions. Complementarily, AI has predictive capabilities. By analyzing historical and real-time patterns, it can anticipate congestion, predict vehicle movements, and proactively adjust packet-forwarding strategies. This synergy not only improves efficiency but also enables the scalability of VANET in complex urban environments.

To illustrate the applications of AI and SDN in VANET routing, Table 9 presents a selection of recent research papers. This table describes the characteristics of each proposal, including the route type, transmission mode, routing algorithm, and other relevant aspects.

Table 10 complements the information in Table 9, showing the performance objective and innovative idea behind each routing protocol or solution in the VANET. In addition, the types of AI and SDN used in each proposal were identified.

Although the integration of AI and SDN in VANET promises significant advances, gaps still exist, which limit their comprehensive implementation. Most studies have focused on improving the performance of specific aspects such as routing and resource management, leaving other challenges such as security, privacy, and quality of service relatively unexplored. This partial approach points to the need for broader comparative analyses evaluating the effectiveness of different SDN architectures and AI implementations in diverse vehicular environments [93], [94], [95], [96], [97], [98].

In addition, the convergence of technologies such as fog computing, blockchain, and AI within SDN-based VANET represents a key opportunity for the development of advanced transportation ecosystems. For example, blockchain can provide robust trust and security mechanisms, whereas AI can improve real-time decision making through techniques such as fuzzy logic and reinforcement learning. However, it is crucial to explore how these technologies can be efficiently integrated into distributed and hierarchical architectures to overcome the limitations inherent in centralized systems [93], [94], [95], [96], [97], [98].

Table 11 summarizes recent research that integrates AI and SDN in the context of VANET, allowing the identification of key advances and current gaps in the implementation of these technologies in vehicular networks, and highlighting their applicability and practical challenges.

VI. VANET AND OTHER TECHNOLOGIES

VANET are not limited to a single technological approach but rather thrive on the integration of diverse technologies to overcome the challenges inherent in their implementation. Although SDN and AI have proven to be valuable tools in this field, other technologies also play important roles. crucial to optimizing the functionality and security of VANET.

An area of interest is the efficient management of the radio frequency spectrum, which is an essential resource for vehicular communications. CR technology has emerged as a promising solution for addressing this challenge. Instead of relying on a statically assigned frequency spectrum such as DSRC, CR enables vehicles to dynamically identify and access unused or sparsely used frequency bands [102]. This approach promises to significantly improve the spectral efficiency and flexibility of VANET [102].

Data security and privacy are critical aspects of VANET, particularly given the increasing amount of sensitive information shared between vehicles. Blockchain technology has emerged as a robust solution for addressing these concerns by providing a secure and transparent mechanism

TABLE 11. Summary of recent research on AI and SDN integration in VANET. Symbols “T”, “S”, “E” and “-” indicate theoretical, simulation, experimental and not mentioned respectively.

Reference	Main focus	AI/SDN application in VANET	Case studies	Specific frameworks	Future applicability
[1]	T/S	Improves performance when managing routing rules	-	Ryu, OpenFlow	Improving the reliability and efficiency of vehicle-to-vehicle communication by creating vehicle clusters.
[54]	T	Predicting vehicle arrival rate and probability of successful transmission. Makes routing decisions between RSUs and base stations	-	-	Minimizing data transmission delay in vehicular networks using mobility prediction to select optimal route
[65]	T/S	Improving vehicle traffic management. Manage traffic and connectivity with flexible and programmable control	-	-	Improving traffic management, road safety, and emergency response through vehicle-to-vehicle communication, infrastructure, and 5G networks.
[67]	T/S	Centralized controller manages network and data dissemination	-	Ns-3, OpenFlow	For vehicles to intelligently select the best available wireless network based on application requirements
[70]	T/S	Selection of the cluster lead vehicle. Improve network management efficiency with controllers at the MEC layer	Demonstrate s speed message dissemination in the proposed architecture	INETmanet-2.0, OpenFlow	Improving the security and reliability of data transmission in vehicular networks using multi-access edge computing and SDN
[72]	T/S	Central controller and RSUC optimize network management	-	OpenFlow	Improving data packet transmission in vehicle-to-vehicle communications by determining the most suitable route using SDN and fog computing
[76]	T/S	Optimizes route selection and efficient packet flow	-	-	Ensuring reliable and low-latency data transmission in VANET applications
[78]	T/S	Centralized control facilitates uninterrupted data transfer	-	-	Improve network performance in intelligent transport systems by reducing packet loss and transmission delay
[81]	T/S	Considering the cost of cellular bandwidth. Manages cluster formation and route selection in VANET	-	-	Collect vehicle data and transmit it to a processing center via a combination of V2V and cellular communication to balance cost and latency.
[82]	T/S	Cloud and cluster controller manage IoV routing	-	Ns-3, OpenFlow	Routing packets safely and efficiently in the Vehicular Internet of Things
[83]	T/S	Central controller authenticates vehicles and generates secure keys	-	OpenFlow	Improving the security and reliability of data transmission in vehicular networks through network encryption and centralized authentication.
[86]	T/S	Combines centralized and distributed routing techniques	-	OpenFlow	Improving data transmission in vehicular networks by selecting the shortest and most stable routes
[87]	T/S	Optimize flow by redirecting traffic based on network load	-	OpenFlow, OFSwitch	Improve the efficiency of vehicular networks by redirecting traffic through less congested nodes
[93]	T/S	Signaling optimization considering environmental factors and vehicle density. Manage data flow and control network devices	Based on intersections and phase designs in Adapazarı, Türkiye	OpenvSwitch, Floodlight	Optimize traffic signals in urban environments, taking into account traffic density and environmental conditions

TABLE 11. (Continued.) Summary of recent research on AI and SDN integration in VANET. Symbols “T”, “S”, “E” and “-” indicate theoretical, simulation, experimental and not mentioned respectively.

[94]	T/S	Optimal link quality policy with malicious vehicles. Learn optimal linking policies with deep Q learning	-	TensorFlow	The quality of communication between vehicles in traffic environments can be improved by intelligently selecting the most reliable and efficient communication routes.
[95]	T/S	Link stability and adaptive area selection. Maintains routing tables and selects connection areas	-	-	Improving data transmission in software-defined vehicle networks
[99]	T/S/E	Distributed SDN controllers manage MEC servers in RAN	Improves traffic flow and facilitates passage for emergency services	OpenFlow	The efficiency, safety, and reliability of transportation systems can be improved by allowing vehicles to offload computing tasks to the edge servers.
[100]	T/S	Optimal computational offloading strategy in vehicles. Dynamically manages security in VANET communications	-	TensorFlow	Improve the safety and efficiency of offloading computing tasks from vehicles to clouds or edge servers.
[101]	T/S	Prediction of vehicle position, route and optimal beam selectio. Manages automatic band switching based on spectrum and QoS	-	OpenFlow	Achieving ultra-high data rates in V2X communications

for authentication, access control, and data integrity protection [103], [104], [105].

Cloud and edge computing offer new possibilities for improving the processing and storage capabilities of VANET. Offloading computationally intensive tasks to the cloud or edge servers allows vehicles to access computing resources on demand, freeing up their local resources and enabling the execution of more complex applications [100], [104], [105]. Another area of interest is MEC, which brings computational resources closer to vehicles, thus reducing latency and improving the efficiency of data offloading [99]. Integrating MEC with other technologies, such as SDN, can further optimize network performance and traffic management [99], and convergence with technologies such as 5G can significantly improve the QoS in VANET [106]. However, data security and privacy remain important considerations, especially when dealing with large amounts of sensitive information such as location data and traffic patterns [99], [107].

Accurate time synchronization is crucial in VANET for various applications such as road safety and traffic management. An Adaptive Beacon Time Synchronization algorithm was proposed in [107] to improve the time synchronization accuracy in large-scale VANET, thereby addressing the challenges posed by high vehicle mobility and dynamic network topologies [107]. Efficient resource management is a critical challenge in VANET. The problem of load balancing in offloading cloud computing tasks in VANET was addressed in [108] by considering the power control and vehicle selfishness.

Advanced communication technologies, such as mmWave and THz, promise to revolutionize vehicular communication by offering ultrafast data rates and extremely low latency, which are essential features for applications such as autonomous driving and road safety [101]. Intrusion detection is essential for ensuring the security and reliability of VANET. A node flow classification method for intrusion detection in VANET was presented in [109], which combined clustering with spline-based general linear model classification [109].

The combination of these technologies with VANET opens a range of possibilities for the creation of smarter, safer, and more efficient transport systems. However, effective integration of these technologies poses significant challenges that must be addressed in future research.

Table 12 presents a range of studies exploring the application of various technologies in VANET beyond SDN and AI.

Fig. 10 shows the various technologies used in VANET-related research. It can be observed that

Blockchain technology is the predominant technology. This suggests that blockchain has become a core technology for developing solutions for VANET, probably owing to its security and decentralization features.

The future of VANET is seen with the convergence of emerging technologies, such as AI, SDN, fog computing, and blockchain, which opens up new opportunities and challenges for smart transportation. These technologies will enable the optimization of various aspects of VANET, from real-time decision-making to the security of transactions between

TABLE 12. VANET and other key technologies.

Reference	Main technology	Aspect of VANET that improves
[99]	Computing in the fog	Latency, network congestion, reliability
[100]	Blockchain	Computer offload security, access control
[101]	Cognitive radio	V2X routing, cluster formation, predictive switching
[102]	Spectrum detection	Spectrum sensing efficiency, radio resource management
[103]	Blockchain	Privacy, Caching Efficiency
[104]	Blockchain	Data security, fog computing, trust management
[105]	Blockchain	Secure message transmission, node reliability prediction
[106]	Network Slicing	Resource allocation, bandwidth management
[107]	ABTS	Time message synchronization, energy efficiency
[108]	Power control	Load balancing, computing offloading
[109]	Node flow classification	Intrusion detection, classification accuracy

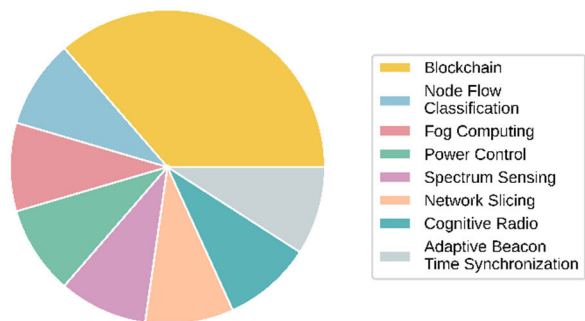


FIGURE 10. Trends in SDN architectures for VANET.

vehicles. Furthermore, the development of distributed and hierarchical SDN architectures, together with the efficient integration of AI, is essential for addressing key challenges such as scalability, resilience, and resource management. The synergy of these technologies can potentially transform transportation systems into safer, more efficient, and more connected ecosystems [99], [100], [103], [104], [105], [107].

VII. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

Despite their potential, large-scale deployment of VANET faces several obstacles owing to the dynamic nature of the network and the complexity of integrating various technologies. To move towards a safer, more efficient, and smarter

ITS, it is essential to address current challenges and explore new avenues of research. The open problems and challenges are described below:

1. Designing distributed AI architectures for VANET environments: The design and creation of AI architectures capable of learning from large amounts of data and adapting to changing VANET conditions. Techniques are required to improve the data quality, protect adversarial attack models, and ensure information privacy.
2. Implementation of federated learning: The AI models used in VANET must continuously learn and adapt to environmental changes. Federated learning allows vehicles to collaborate in updating the models.
3. Optimizing SDN for vehicular environments: SDN offers great potential for efficiently managing vehicular networks. However, it must be adapted to the dynamic characteristics of such networks. This involves the development of more resilient architectures, adaptive routing protocols, and efficient resource-management mechanisms.
4. AI and SDN integration: Combining AI and SDN can revolutionize VANET management, enabling smarter and more autonomous decision making. However, further research is required to effectively integrate these technologies and develop algorithms that enable their collaboration.
5. Efficient resource management in highly mobile environments: Allocation of resources, such as bandwidth and transmission power, is a challenge in VANET because of the constant mobility of vehicles. It is necessary to develop resource management algorithms that can adapt to the changing demands of a network and optimize its utilization.
6. Security and privacy in heterogeneous environments: The increasing complexity of VANET poses new challenges in terms of security and privacy, owing to the integration of multiple technologies. Therefore, it is essential to develop robust protection mechanisms to ensure data confidentiality and to prevent attacks.

Table 13 presents the research directions, problems investigated, recommended solutions, and suggested references used to complement this study.

Although the integration of AI and SDN in VANET promises significant advances, gaps still exist, which limit their comprehensive implementation. Most studies have focused on improving the performance of specific aspects, such as routing and resource management, leaving other challenges such as security, privacy, and quality of service relatively unexplored. This partial approach points to the need for broader comparative analyses that evaluate the effectiveness of different SDN architectures and AI implementations in diverse vehicular environments [93], [94], [95], [96], [97], [98].

TABLE 13. Challenges of open research in VANET with AI, SDN and other technologies.

Challenge	Problem	Solution proposal	Reference
Limited data sets for AI	Insufficient and low-diversity data sets	Create larger sets, use synthetic data	[40], [59], [60]
Robustness and resistance of AI to attacks	Risk of AI models being manipulated or compromised	Develop defense mechanisms and federated learning	[52], [53], [54], [55], [56], [57]
Dynamics and distribution of VANET for AI	Adapting algorithms to changing and distributed environments	Efficient algorithms, distributed architectures	[44], [45], [46]
Security and privacy in VANET with AI	Risk of exposure of sensitive information	Anonymization, privacy-preserving learning, access control	[24], [56], [58], [59]
Unstable connectivity in VANET with SDN	Fluctuations in network topology	Distributed SDN architectures, adaptive routing protocols	[65], [67], [71], [82]
Security in VANET with SDN	Risk of attacks on SDN centralization	Strong authentication, cryptography, intrusion detection systems	[41], [42], [47], [63]
Efficient resource management in VANET with SDN	Dynamic resource allocation in a dynamic environment	Adaptive resource management algorithms, machine learning	[87], [88], [89]
SDN scalability in VANET	Managing vehicle and data growth	Scalable SDN architectures, network functions virtualization	[41], [42]
Complexity of SDN and AI integration in VANET	Complex management of integrated systems	Management, orchestration and automation frameworks and tools	[72], [74], [86]
Optimizing synergy between SDN and AI in VANET	Finding the best combination of SDN and AI	New algorithms and strategies, reinforcement learning	[66], [71], [72], [74]
Continuous learning in AI for VANET with SDN	Continuous updating of AI models	Online machine learning, federated learning	[72], [74], [86]
Interoperability between different systems and technologies in VANET	Lack of standards and protocols for integrating blockchain with technologies such as SDN, AI and	Common standards and protocols, open architectures	[34], [93], [94]

TABLE 13. (Continued.) Challenges of open research in VANET with AI, SDN and other technologies.

	traffic management systems.		
Scalability and Resource Management	Difficulty scaling blockchain on VANET due to high resource consumption	Developing lightweight blockchain solutions	[21], [57], [64], [77], [88], [89]
Adapting blockchain to the dynamic architecture of VANET	Changing topology and mobile nodes make it difficult to implement rigid consensus protocols.	Design flexible consensus protocols and efficient information management for networks with high mobility.	[6], [7], [8], [9], [10], [11], [41], [42]
Enhanced security and privacy for large volumes of data	Protecting sensitive information and preventing attacks	Implement protection mechanisms for data privacy and security in blockchain for VANET.	[1], [42], [51], [61]

In addition, the convergence of technologies such as fog computing, blockchain, and AI within SDN-based VANET represents a key opportunity for the development of advanced transportation ecosystems. For example, blockchain can provide robust trust and security mechanisms, whereas AI can improve real-time decision making through techniques such as fuzzy logic and reinforcement learning. However, it is crucial to explore how these technologies can be efficiently integrated into distributed and hierarchical architectures to overcome the limitations inherent in centralized systems [93], [94], [95], [96], [97], [98].

VIII. DISCUSSIONS

Below, some research gaps detected in this research work will be highlighted:

A. SCALABILITY AND RELIABILITY IN DYNAMIC VANET ENVIRONMENTS

VANET operate in highly dynamic environments characterized by rapid vehicle mobility, fluctuating traffic density, and inconsistent connectivity, which pose significant challenges for scalability and reliability. SDN offers benefits, such as centralized control and network programmability, and its application in VANET has inherent limitations. Centralized SDN controllers, which are efficient in static networks, struggle with the constantly changing topology of VANET, leading to latency in routing decisions, owing to the time required for data collection and instruction dissemination. This delay can critically impact real-time applications, such as collision avoidance systems.

To address these challenges, hybrid or distributed SDN architectures have been proposed, where control is divided

among multiple local controllers, reducing reliance on a single point of failure and enabling quicker decision making for tasks such as short-route management. Local controllers can manage immediate adjustments in dense urban areas, whereas central controllers coordinate broader strategies. Integrating edge computing and blockchain technologies can further enhance adaptability and resilience. Edge computing reduces latency by processing data closer to the source, and the blockchain can establish distributed consensus systems for route validation without depending on a central authority. These innovations aim to create a dynamic and responsive network capable of adapting to the unique demands of VANET.

B. SECURITY AND PRIVACY IN VEHICULAR COMMUNICATION

VANET handle sensitive data, such as vehicle locations and identifiers, making them prime targets for cyberattacks, such as spoofing and data interception. Although advancements in artificial intelligence have improved intrusion detection systems, significant challenges remain, particularly in balancing privacy and computational efficiency. Federated learning offers a promising approach by enabling collaborative AI model training without sharing raw data, thereby preserving privacy. However, its implementation in VANET is complicated by device heterogeneity because many vehicular communication modules lack the computational power to run resource-intensive federated learning models. Additionally, inference risks persist where attackers may reconstruct sensitive information, such as movement patterns, from shared model updates.

To address these issues, combining federated learning with techniques such as differential privacy or homomorphic cryptography can enhance security by preventing the central servers from accessing sensitive data. Lightweight encryption methods are also critical, as traditional encryption standards like AES are resource-heavy for IoT-enabled vehicles. Alternatives, such as identity-based encryption (IBE), pseudonymization techniques, and secure hardware solutions, such as Trusted Platform Modules (TPMs) offer practical ways to protect data while minimizing computational overhead. IBE simplifies key management; pseudonyms help avoid tracking through frequent identifier rotation; and TPMs provide isolated storage for cryptographic keys, ensuring robust yet efficient security for VANET.

C. OPTIMIZING THE INTEGRATION BETWEEN AI AND SDN IN VANET

The integration of AI and SDN into VANET aims to create adaptive, efficient, and resilient networks. However, this convergence faces significant challenges, particularly when addressing the dynamic and unpredictable nature of vehicular environments. Current AI models, often trained on historical data, struggle to respond effectively to real-time disruptions, such as accidents, natural disasters, or sudden traffic changes. Reinforcement learning offers a potential solution

by enabling systems to adapt dynamically through immediate feedback from the network, improving routing and resource allocation, while reducing latency and packet loss.

Achieving seamless coordination between SDN and AI is another critical challenge. For optimal performance, SDN should serve as the source of global network data, whereas AI acts as a decision-making engine that processes these data to generate actionable insights. This synergy allows for real-time adjustments and intelligent decision making, ensuring that VANET can respond effectively to changing conditions. By aligning the strengths of both technologies, AI-driven SDN architectures can enhance the overall efficiency and reliability of vehicular networks, paving the way for smart transportation systems.

IX. CONCLUSION

VANET are undergoing a significant transformation owing to the integration of emerging technologies such as AI and SDN. It has been demonstrated that these technologies enable a new paradigm in ITS, fostering safer, more efficient, and connected mobility. First, an in-depth literature review on VANET is provided, including their interrelation with IoV and other relevant technologies, such as edge computing, AI, SDN and Blockchain, analyzing the key approaches and contributions of different studies in these areas and highlighting the relationships between the topics addressed. Next, this paper focuses on the AI techniques used in VANET, including ML and deep learning, and how they can improve transportation safety and efficiency.

The challenges and limitations of applying AI to VANET are discussed, including the need for large-scale datasets and the robustness of algorithms against attacks. AI, through techniques such as ML and deep learning, enables the extraction of valuable insights from large volumes of data generated by vehicles, thereby improving real-time decision-making and anomaly detection. The impact of SDN on VANET management and optimization was then analyzed, highlighting its advantages in terms of centralized control, flexibility, and scalability and explaining how SDN can improve security in VANET by enabling centralized control of security policies and faster response to threats. SDN offers greater flexibility and programmability in VANET management, facilitating the implementation of security policies and routing optimization. The synergy between AI and SDN in VANET was then explored, showing how AI can enhance SDN capabilities by enabling intelligent decision making and dynamic adaptation to network conditions. The convergence of AI and SDN promises a future in which vehicles can communicate intelligently and adapt to changing traffic conditions, thereby improving the road safety and energy efficiency. The importance of integrating VANET with other emerging technologies, including CR, Blockchain, Cloud Computing, Edge Computing, and advanced communication technologies, is also discussed as these offer opportunities to further improve VANET.

Finally, the main challenges and limitations in deploying VANET on a large scale were analyzed, identifying key areas that require further research. However, several challenges have been identified that require future research, such as the need to develop more robust and adaptable AI models, ensure data privacy, and optimize resource management in highly dynamic environments. This review provides a comprehensive perspective on the current state and future trends of VANET, focusing on the effects of AI, SDN, and other emerging technologies on their development. This paper lays a solid foundation for future research in the field of VANET, as it not only summarizes the current state of the technology but also identifies the limitations of current solutions and proposes directions for future research.

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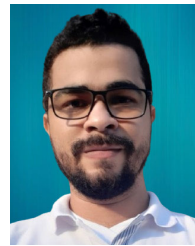
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